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2025-yil 2-variant

1. a sonining 36%i, 108 ning 24% iga teng. a sonini toping.

A) 36
B)108
C)72
D)54

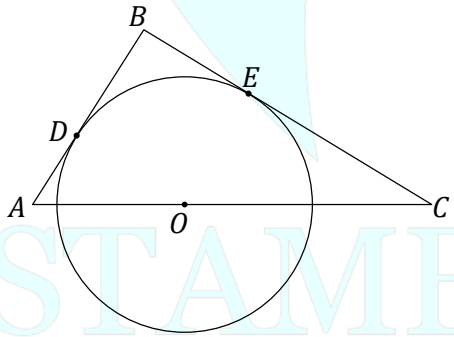
2. Agar $f(\log_3 x - 1) = x^2 - 8x + 3$ bo'lsa, $f'(1)$ ni toping.

A)$90ln3$
B) $6ln3$
C)$27ln3$
D)$\frac{10ln3}{9}$

3. Agar $f(\log_5 x - 1) = x^2 - 8x + 3$ bo'lsa, $f'(1)$ ni toping.

A)$100ln3$
B)$10ln3$
C)$1050ln5$
D)$\frac{100ln3}{9}$

4. ABC uchburchakning $AB = 13$, $BC = 15$ va $AC = 14$ AC tomoni aylana markazidan qolgan tomonlari AB va BC tomonlari esa aylanani D va E nuqtalarda urinsa, AD kesma uzunligini toping.

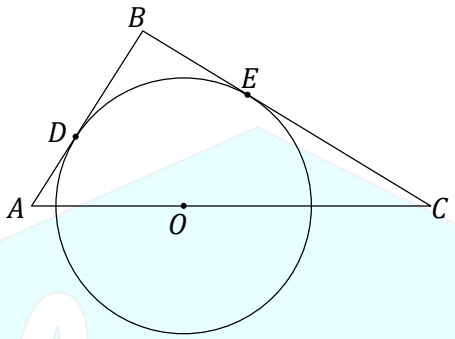


A) 1
B) $\frac{7}{9}$
C) $\frac{5}{3}$
D)$\frac{3}{2}$

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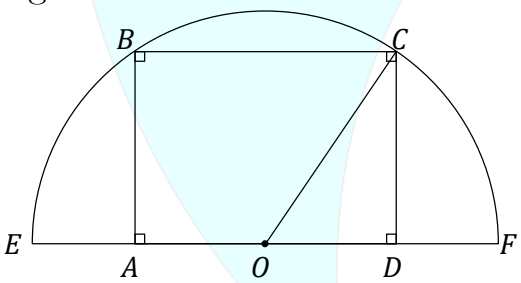
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5. ABC uchburchakning $AB = 12$, $BC = 15$ va $AC = 14$ AC tomoni aylana markazidan qolgan tomonlari AB va BC tomonlari esa aylanani D va E nuqtalarda urinsa, AD kesma uzunligini toping.



A) $\frac{116}{57}$
B) $\frac{113}{54}$
C) $\frac{115}{54}$
D) $\frac{5}{3}$

6. R radiusli yarim doiraga ichki chizilgan eng katta yuzaga ega bo'lgan to'g'ri to'rtburchakning yuzini toping.



A) R^2
B) $\frac{R^2}{3}$
C) $\frac{R^2}{2}$
D) $\frac{R^2}{4}$

7. Agar $tgx + ctgx = a$ bo'lsa, $tg^5x + ctg^5x$ ni a orqali ifodalang?
A) $a^5 - 5a^3 + 5a$
B) $a^5 + 4a^3 - 2a$
C) $a^5 - 3a^3 + 2a$
D) $a^5 - 4a^3 + 2a$

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8. Tenglamaning $8cos\frac{3x}{2}\cdot cos\frac{x}{2}-3=6cos2x$ eng katta manfiy yechimini toping?

A) $-\frac{\pi}{2}$

B) $-\frac{\pi}{6}$

C) $-\frac{\pi}{4}$

D) $-\frac{\pi}{3}$

9. Agar $\sqrt{sina-\sqrt{sina-\sqrt{sina-\dots}}}=\frac{1}{2}$ bo'lsa, $sina$ ning qiymatini toping.

A) $\frac{1}{2}$

B) $\frac{3}{4}$

C) $\frac{1}{4}$

D) $\frac{2}{3}$

10. Aniqmas integralni hisoblang: $\int x^8 ln8x\,dx$

C) $\frac{x^8 ln8x}{9}-\frac{x^9}{81}+C$

B) $\frac{8x^7 ln8x}{9}-\frac{x^9}{81}+C$

C) $\frac{x^9 ln8x}{9}-\frac{x^8}{81}+C$

D) $\frac{8x^7 ln8x}{9}-\frac{x^8}{81}+C$

11. Agar $f(x)=(1+x^2)\cdot\sqrt{2+x^2}\cdot\sqrt[3]{2+x^2}$ bo'lsa $f'(1)=?$

A) $\frac{28\sqrt[6]{243}}{9}$

B) $\frac{28}{\sqrt[6]{3^8}}$

C) $\frac{28}{\sqrt[6]{3^9}}$

D) 1

12. Tenglamaning $\left[-\frac{\pi}{2};\frac{\pi}{2}\right]$ oraliqdagi yechimlari sonini toping.

$$cos^2 6x+sin^2 3x+3sin3x+2=3cos6x+2sin3x\cdot cos6x$$

A) 5

B) 6

C) 4

D) 7

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13. Ikkita sonning yig'indisi 10 ga teng, bu sonlarning o'rta geometrigi 4 ga teng. Bu sonlar ayirmasining modulini toping.

A) 3
B) 4
C) 5
D) 6

14. Aniq integralni hisoblang: $\int_0^3 \sqrt{9-x^2}dx$

A) 3π
B) $\frac{7\pi}{4}$
C) 2π
D) $\frac{9\pi}{4}$

15. Agar a, b ratsional sonlar uchun $a+2\sqrt{2}b=12$ tenglik o'rinli bo'lsa, $a+b$ ning qiymatini toping.

A) 12
B) 13
C) 14
D) 18

16. Agar $sina=0{,}6$ bo'lsa, $a \in \left(0;\frac{\pi}{2}\right)$ oraliqda bo'lsa $ctg\left(\frac{\pi}{2}-2a\right)$ni toping.

A) $\frac{25}{7}$
B) $\frac{24}{7}$
C) $\frac{12}{7}$
D) $\frac{20}{7}$

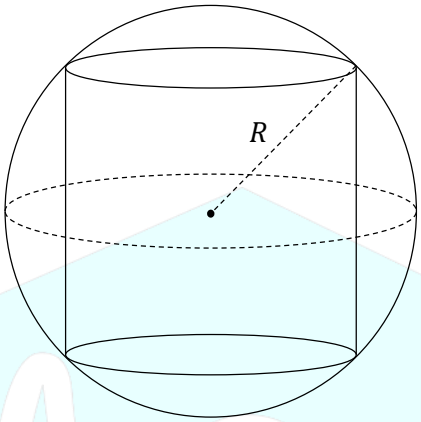
17. Aniq integralni hisoblang: $\int_1^2 (x-1)\left(x^4+x^3+4x^2+\frac{x+1}{x}+8\right)dx$

A) $\frac{215}{12}-ln2$
B) $\frac{219}{12}-ln2$
C) $\frac{97}{12}-ln3$
D) $\frac{97}{12}+ln2$

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18. Sharga eng katta hajmli silindr ichki solingan. Agar shar radiusi $R = \sqrt{3}\ sm$ bo'lsa silindrning hajmini toping?



A)8π
B)6π
C)$6\sqrt{6}\pi$
D)4π

19. Hisoblang:

$$\frac{ln16}{ln10 - ln5}$$

A)2
B) 4
C)6
D)8

20. Hisoblang

$$\frac{1}{2+\sqrt{3}} + \frac{1}{2-\sqrt{3}}$$

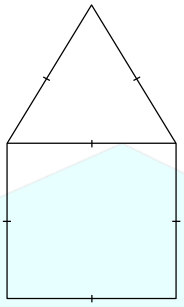
A)4
B)3
C)$2+\sqrt{3}$
D)1

21. 14 ga bo'linadigan ikki xonali sonlar nechta?
A) 6 ta
B) 8 ta
C) 9 ta
D) 7 ta

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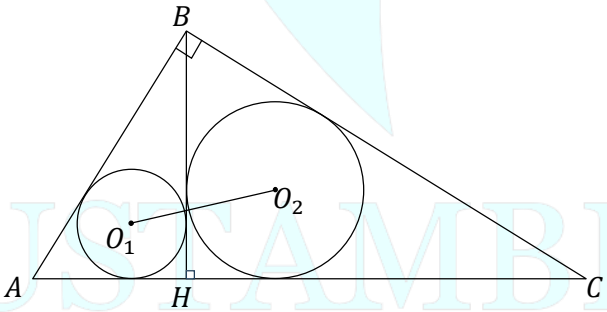
22. Rasmda muntazam uchburchak va kvadrat berilgan. Kvardratning yuzi 36 ga teng bo'lsa, uchburchakning perimetrini toping.



A)18
B)16
C)24
D)12

23. Tenglamaning haqiqiy ildizlari nechta: $x^2 - 5|x| + 6 = 0$
A)4ta
B)3ta
C)2ta
D)5ta

24. Tomonlari $AB = 12$, $BC = 16$, $AC = 20$ bo'lgan uchburchakka BH balandlik tushirilgan. ABH va BHC uchburchaklarga ichki chizilgan aylanalar markazlari orasidagi masofani toping $(O_1O_2 = ?)$.



A)$8\sqrt{2}$
B)$8\sqrt{2}$
C)$4\sqrt{2}$
D)4

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25. a, b natural sonlar uchun $a + \frac{b}{2} = 100$ tenglik o'rinli bo'lsa $a + b$ ning eng katta qiymatinig toping.
A)200
B)250
C)300
D)199

26. Agar $\alpha, \beta \in \left(0; \frac{\pi}{2}\right)$ va $\operatorname{tg}\alpha = 3$, $\sin\beta = \frac{2\sqrt{2}}{3}$ bo'lsa, $\sin 2\alpha + \cos\beta = ?$
A)$\frac{\sqrt{10}}{30}$
B) 1
C) $\frac{14}{15}$
D) $\frac{13}{15}$

27. Muntazam to'rtburchakli piramida hajmi 288 ga teng. Piramidaning balandligi asosining diagonaliga teng bo'lsa, piramida yon sirtini toping.
A)256
B)192
C)108
D)216

28. $2ax - 2ay + 2bx - 2by$ ni soddalashtiring.
A)$(x - y)(2a + b)$
B)$(x - y)(a + 2b)$
C)$2(x + y)(a + b)$
D)$2(x - y)(a + b)$

29. Gul do'konida qizil atirguldan 3 xil guldasta va oq atirguldan 2 xil guldasta yasalgan. Bu guldastalardan 1 tasini olishning nechta usuli bor?
A)4
B) 5
C)9
D)12

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30. Trapetsiya diagonallari 9 va 12 ga teng bo'lib ular o'zaro perpendikulyar bo'lsa, trapetsiya o'rta chiziq uzunligini toping.

A)5
B)6
C)4
D)7,5

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